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Git-Link:- [https://github.com/Mayank-dub/AnalogElectronicsLab\\_2025BEC0029/tree/main/Eperiment04\\_current\\_mirror](https://github.com/Mayank-dub/AnalogElectronicsLab_2025BEC0029/tree/main/Eperiment04_current_mirror)

AIM: - to create an simple current mirror having  $gm/I_d = 15$  and a reference current of 1mA

To create a cascoded current mirror drawing 2mA of current and  $gm/I_d=15$ .

Procedure:- Necessary derivations and calculations are as follows:-

### Electrical Parameters and Sizing Calculations for $gm/I_d = 15 \text{ V}^{-1}$

#### 1. Overdrive Voltage ( $V_{ov} = V_{GS} - V_{th}$ ):

- Formula:  $gm / I_D \approx 2 / V_{ov}$
- Calculation:  $V_{ov} \approx 2 / 15 \approx \mathbf{0.133 \text{ V (133 mV)}}$

#### 2. Gate-to-Source Voltage ( $V_{GS}$ ):

- In gpdk090, nominal threshold voltage  $V_{th0} \approx 0.28 \text{ V}$  to  $0.30 \text{ V}$ .
- Therefore, the expected gate bias voltage is:  $V_{GS} = V_{th} + V_{ov} \approx 0.30 \text{ V} + 0.133 \text{ V} \approx \mathbf{0.43 \text{ V to } 0.45 \text{ V}}$

#### 3. Current Density ( $J_D = I_D / W$ ):

- From the lab characterization plot at  $gm / I_D = 15 \text{ V}^{-1}$ :  $(I_D / W) \approx \mathbf{10 \mu A/\mu m}$

#### 4. Transistor Width Sizing (at $L = 200 \text{ nm}$ ):

- **For 1 mA Design ( $I_D = 1000 \mu A$ ):**  $W(1mA) = I_D / (I_D / W) = 1000 \mu A / (10 \mu A/\mu m) = \mathbf{100 \mu m}$  (Implementation:  $W = 5 \mu m$  with multiplier  $m = 20$ , or 20 fingers)
- **For 2 mA Design ( $I_D = 2000 \mu A$ ):**  $W(2mA) = I_D / (I_D / W) = 2000 \mu A / (10 \mu A/\mu m) = \mathbf{200 \mu m}$  (Implementation:  $W = 5 \mu m$  with multiplier  $m = 40$ , or 40 fingers)

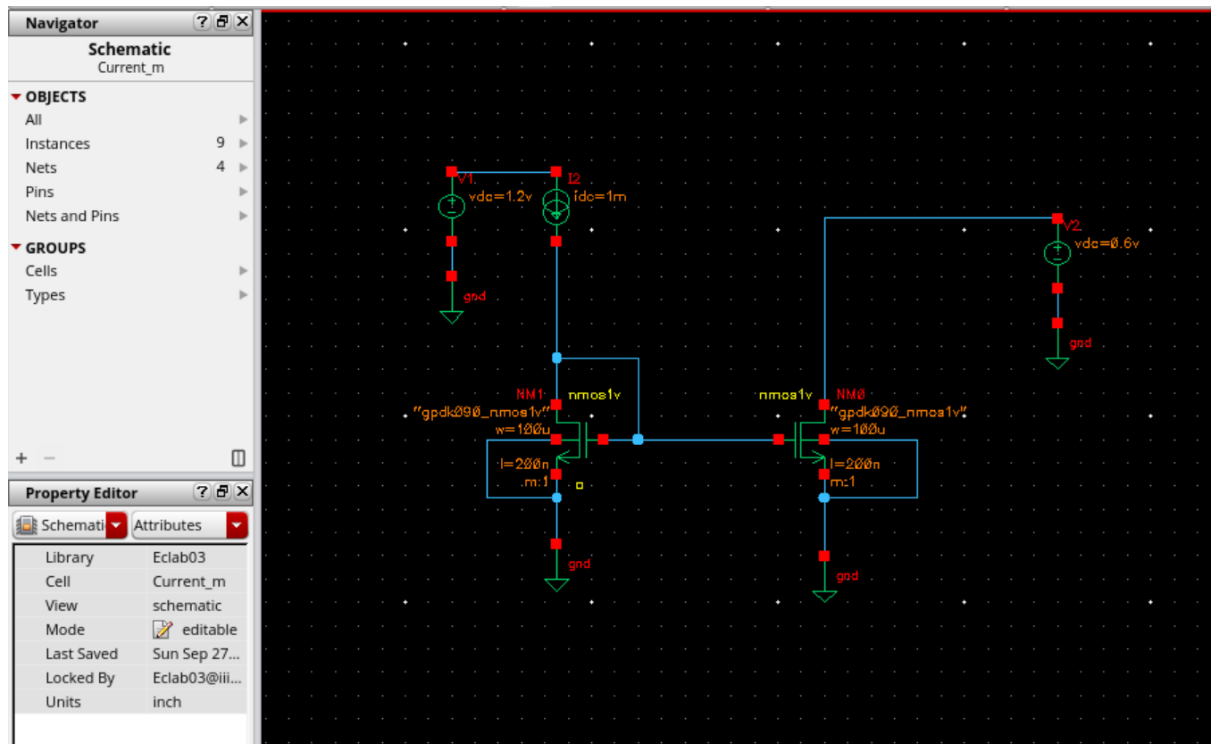
## 5. Small-Signal Transconductance ( $g_m$ ):

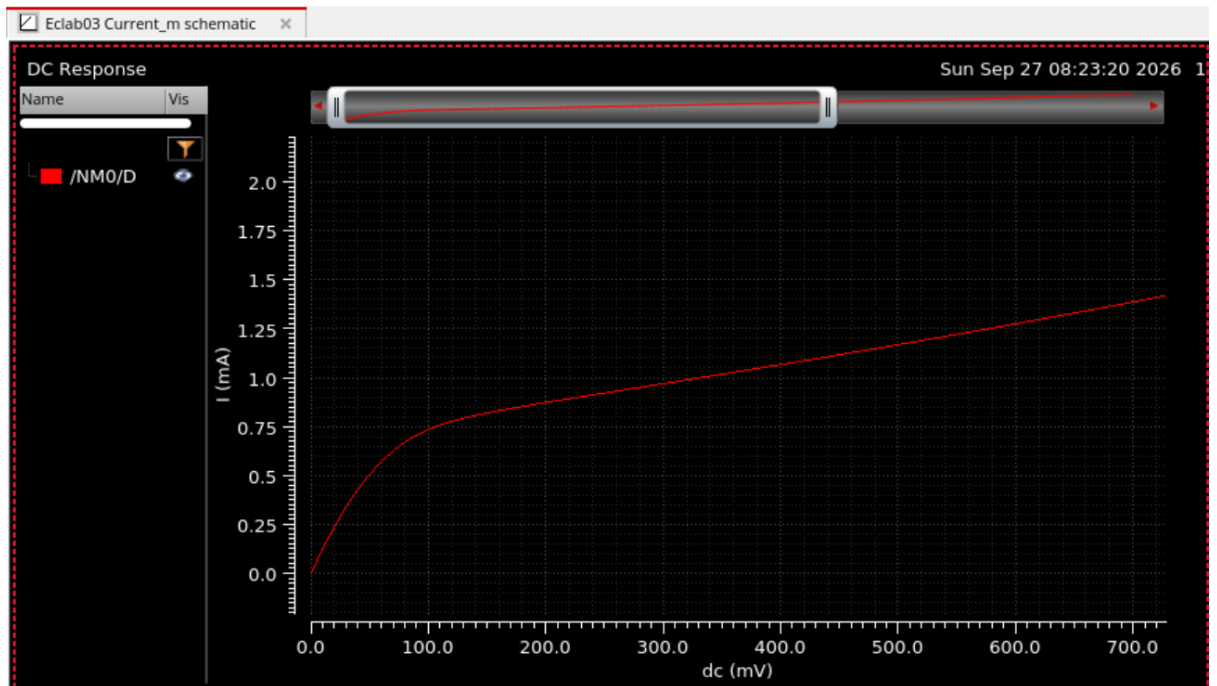
- For 1 mA:  $g_m = 15 \times 1 \text{ mA} = \mathbf{15 \text{ mS (15 mA/V)}}$
- For 2 mA:  $g_m = 15 \times 2 \text{ mA} = \mathbf{30 \text{ mS (30 mA/V)}}$

## 6. Minimum Output Compliance / Knee Voltage ( $V_{out,min}$ ):

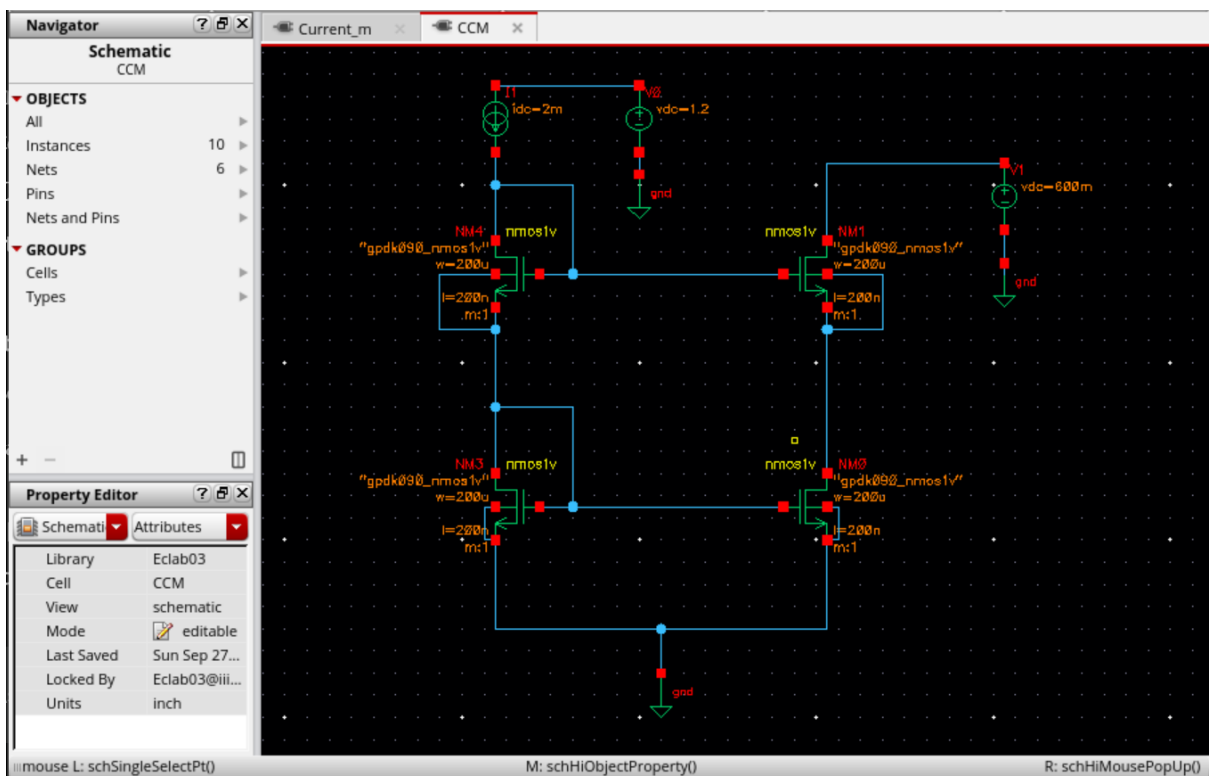
- **Simple Current Mirror:**  $V_{out,min} \approx V_{ov} \approx \mathbf{0.133 \text{ V}}$
- **Cascode Current Mirror:**  $V_{out,min} \approx V_{th} + 2 \times V_{ov} \approx 0.30 \text{ V} + 2(0.133 \text{ V}) \approx \mathbf{0.56 \text{ V to } 0.60 \text{ V}}$

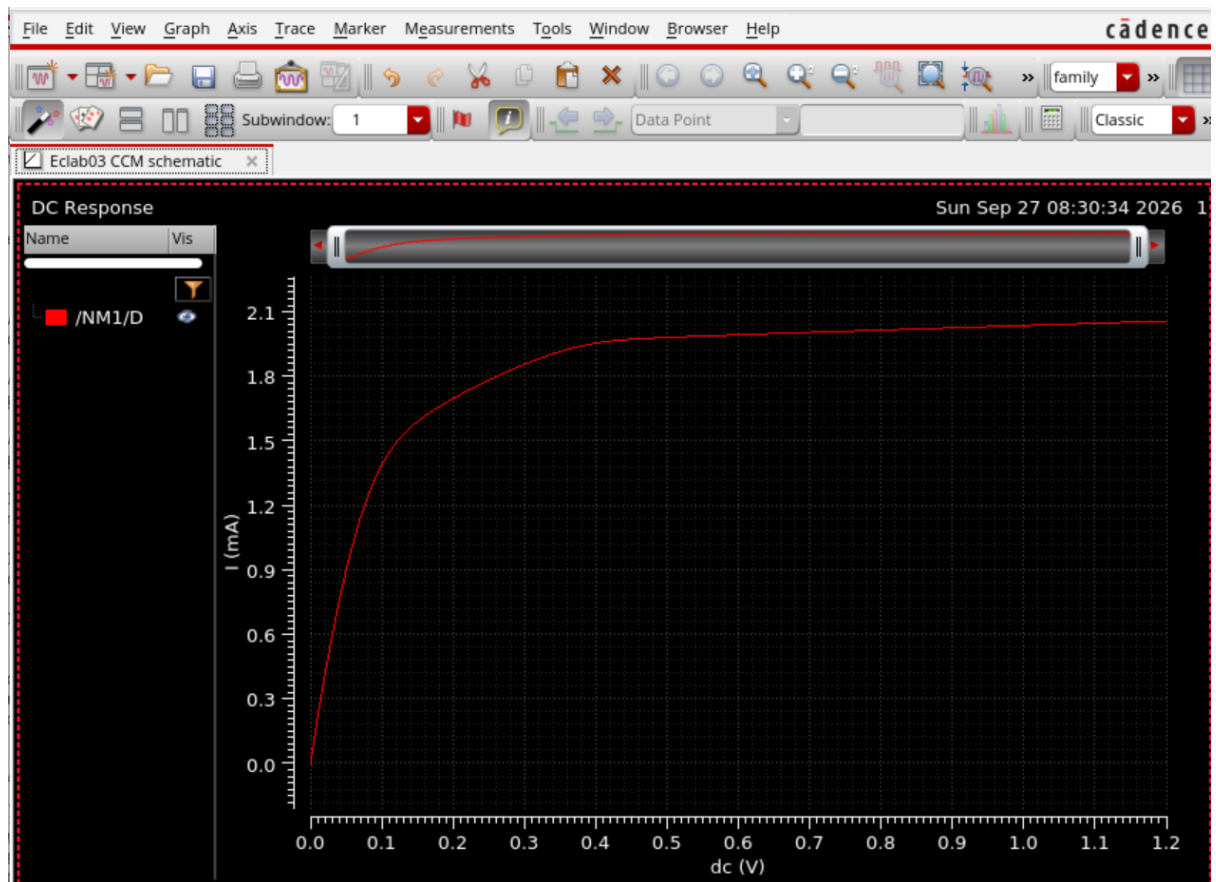
### 1:- Simple current mirror:-





## 2:- Cascoded current mirror:-





## Reason for a flatter saturation region in case of Cascoded current mirror:-

- **Output Resistance ( $R_{out}$ ):** Changes from a single transistor value ( $r_o$ ) to  $g_m \cdot r_o^2$ , acting like a near-infinite barrier to voltage changes.
- **Saturation Slope ( $\Delta I / \Delta V$ ):** Becomes  $1 / (g_m \cdot r_o^2) \approx 0$ , which causes the curve to look completely flat.
- **Top Transistor Drain Voltage ( $V_{D2} / V_{out}$ ):** Acts as the punching bag, absorbing almost all external voltage swings.
- **Intermediate Node Voltage ( $V_{DS1}$ ):** Becomes pinned / frozen at a constant voltage ( $V_{G,cascode} - V_{GS2}$ ).
- **Bottom Transistor:** Acts as a pure, shielded current source, immune to what is happening at the output rail.
- **Top Transistor Gate-to-Source Voltage ( $V_{GS2}$ ):** Acts as an automatic brake (negative feedback)—if current tries to rise,  $V_{GS2}$  drops and immediately pulls it back down.

- **Channel-Length Modulation ( $\lambda$  effect):** Becomes neutralized, preventing the current from drifting upward with output voltage